$$\lambda = \frac{\sin\theta}{Nm} \qquad (7.51)$$

Here, N is the number of rulings per metre in the grating and m is the order of the diffraction image.



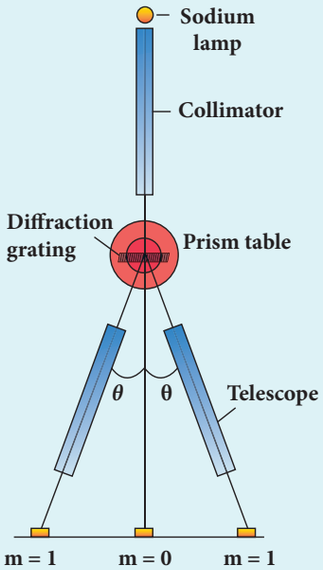


Figure 7.21 Determination of wavelength using grating

7.4.8 Determination of wavelength of different colours

The diffraction pattern for white light consists of a white central maximum and continuous coloured diffraction pattern on its both sides. The central maximum is white as all the colours constructively meet at centre with no path difference. As θ increases, the path difference fullfills the condition for maxima of different orders for all colours from violet to red. It produces a spectrum of diffraction pattern from violet to red on either side of central maximum as shown in Figure 7.22. By measuring the angle at which these colours appear for various orders of diffraction, the wavelength of different colours could be calculated using the formula given by equation (7.51),

$$\lambda = \frac{\sin\theta}{Nm}$$

Here, N is the number of rulings per metre in the grating and m is the order of the diffraction image.

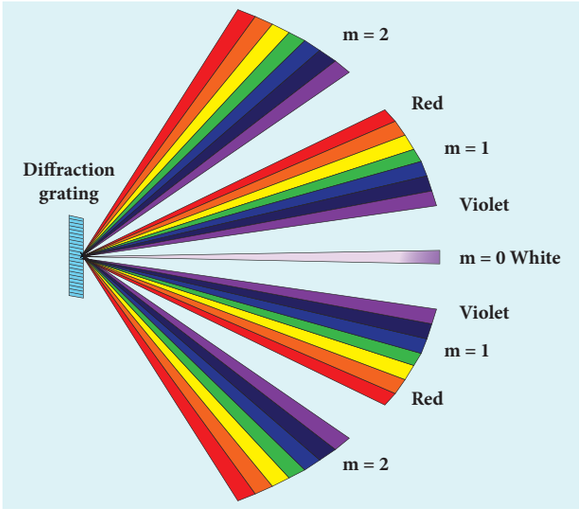


Figure 7.22 Diffraction with white light

7.4.9 Resolution

The effect of diffraction has an adverse effect in the sharpness of the image formed. There is always a spread of central maximum in the image for every point of the object, for every point of the object acts as a point source. The condition for central maximum (or first minimum) produced by rectangular slit is given by the equation (7.37).

$a \sin\theta = \lambda$

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